

Assignment on Securing Accounts for CS50 Cybersecurity

? 10/10 إجمالي النقاط

All assignments in CS50 Cybersecurity are out of 10 points. A score of 7 points or better (70%) is required to be considered to have "passed" an assignment in this course. **Please do not resubmit an assignment if you have already obtained a passing score**—we consider that spam, and if detected, the submission will be deleted, meaning you will not receive the score back anyway. You don't receive a final grade at the end of the course, so it will have no bearing on your certificate, and it will only slow down our graders!

Submitting this assignment more than twice in a single grading cycle constitutes spamming the staff, as well. Upon your third submission, all of your submissions from the current grading cycle are automatically deleted from the system. You will, therefore, not receive any feedback and will need to wait at least three weeks before you submit again.

Unlike CS50x, assignments in this course are graded on a set schedule, and depending on when you submitted, it may take up to three weeks for your work to be graded. Do be patient! Project scores and assignment status on cs50.me/cybersecurity (e.g. "Your submission has been received...") will likely change over time and are not final until the scores have been released.

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* ?What is your GitHub username

If you do not already have a GitHub account, you can sign up for one at <https://github.com/join>. You can then use this account to log in to cs50.me/cybersecurity to track your progress in the course (your progress will only show up after you have received at least one score release email from CS50 Bot, so do be patient!). Don't worry about seeing a 'No Submissions' message on submit.cs50.io, if you find that. The course collects submissions using Google Forms, and only the gradebook on cs50.me/cybersecurity is important

Be certain the username you provide is correct! If you provide the wrong username, •
•.you will not be able to see your scores

Your GitHub username should not be changed while you are taking this course. The •
•.current gradebook system is not designed to accommodate name changes

Be sure to remove extraneous characters, such as an @ prefix. Do not input a URL or •
•.email address, just your username

imaneTech

* City, State, Country

Morocco

* Acknowledgement

Unlike our course CS50x, grading in this course is not done automatically, and there are human reviewers for each assignment. Grading may, depending on exactly when in our grading cycle you submit, take up to three weeks from the time you submit. Your grade status may change in your gradebook at cs50.me/cs50cy in the interim, but grades are never final until you receive a score release email from CS50 Bot (on this first assignment, in fact, your gradebook may not even become active until that score release email). The staff cannot entertain requests for expedited grading under any circumstance. Your patience is appreciated

I understand. ☒

* This course is graded by human graders, and has a ZERO TOLERANCE plagiarism and collaboration policy. If *any* of your answers are copied and pasted from, or obviously based on (a) an online source, including non-course-sanctioned generative AI tools or (b) another student's work in the course, in *any* of the course's five assignments or the final project, you will be reported to edX and removed from the course immediately. There is no opportunity for .appeal. There are no warnings or second chances. It is far better, we assure you, to leave an answer blank rather than risk it. This may be an online course, but it is offered by Harvard, and we're going to hold you to that .standard. **The full essence of all work you submit to this course should be your own**

I understand this policy and agree to its terms; I hereby affirm that I will not plagiarize any answers in this course.



1/1 Why might being required to change our passwords regularly actually pose ✓
?a threat to our security

Changing passwords too often can hurt security because people tend to choose weak, easy-
.to-remember passwords or reuse patterns, making them easier for hackers to guess

1/1 If I have a six-character password consisting of uppercase (English) letters ✓
and (decimal) digits **only**, how many seconds might it take an adversary to
?crack, assuming they make one attempt per second



2176782336



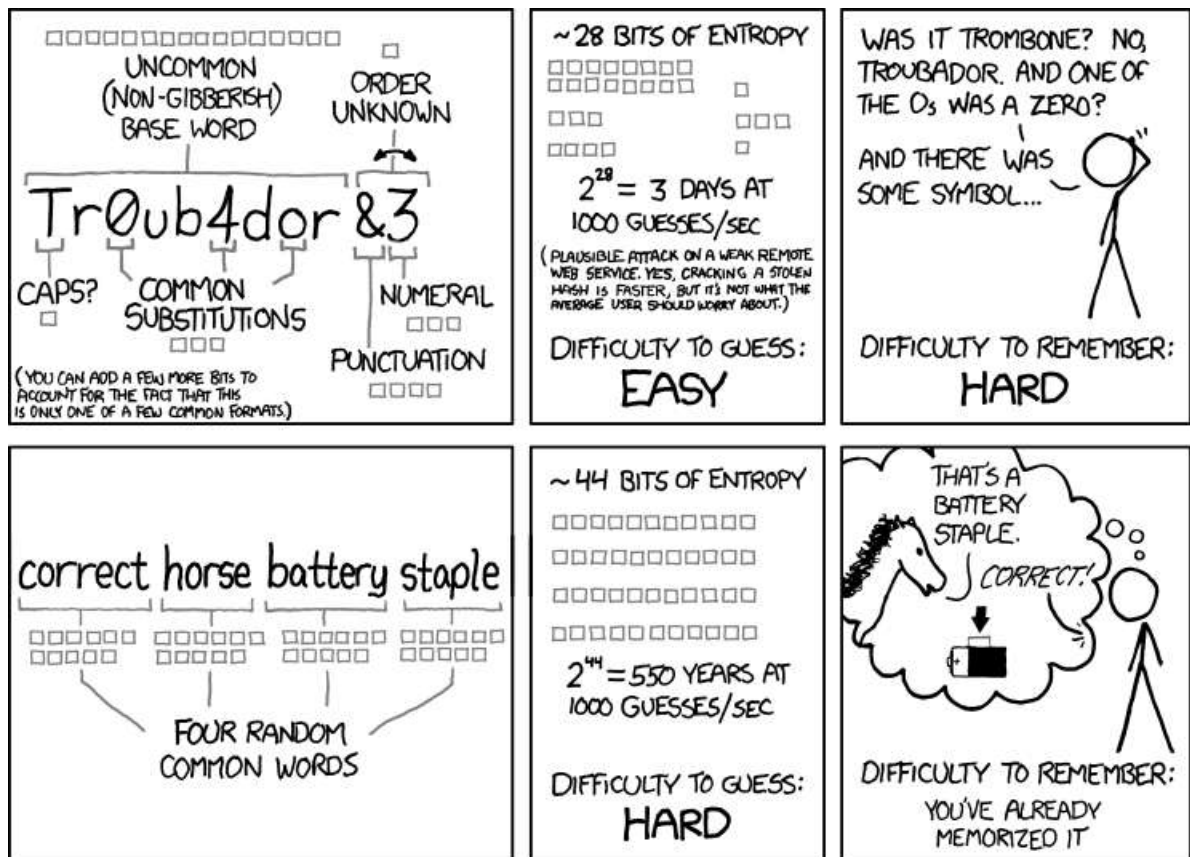
Humor us for a moment, and play [The Password Game](#), trying to get ✓
 .through at least Rule 12

While obviously the game itself is in many ways meant to be humorous, it also critiques the experience many of us have setting up new passwords.

Explain how there's a trade-off between usability and security in the .context of passwords

The Password Game is funny because it shows how password rules can become so complicated that they confuse users instead of helping them. It's a joke about how hard it is .to balance security and usability

Consider the below comic for the next two questions.



THROUGH 20 YEARS OF EFFORT, WE'VE SUCCESSFULLY TRAINED EVERYONE TO USE PASSWORDS THAT ARE HARD FOR HUMANS TO REMEMBER, BUT EASY FOR COMPUTERS TO GUESS.

1/1 Consider the top row of the comic above. Why are passwords like those ✓
?easy (for a computer) to guess but hard (for a human) to remember

Because they follow common patterns (like capital letters, number substitutions, and symbols), which make them predictable for computers to guess. But they are hard for humans to remember because the substitutions and structure don't form anything .meaningful

1/1 Now consider the bottom row of the comic above. Why are passwords like ✓
?those hard (for a computer) to guess but easy (for a human) to remember

Because they use random, unrelated common words, which create a large number of possible combinations — making them hard for computers to crack. But they're easy for .humans to remember because they form a simple, silly phrase

1/1 ?What is a "credential stuffing" attack ✓

A credential stuffing attack is when attackers use stolen usernames and passwords from .one website to try to log in to other websites

1/1 Provide a specific example of something that would be considered a **type** ✓
.of **knowledge factor** for authentication purposes

Do NOT provide any of your own knowledge factors (or anything resembling them) themselves as an answer to this question. We are looking for you to answer the question .in the general sense (a "type of")

If you provide an answer that is, or appears to be, an actual specific knowledge factor, the answer WILL be marked incorrect, without exception, and you should consider that .knowledge factor to have been compromised

✗

password

1/1 Provide a specific example of something that would be considered a **type** ✓
.of inherence factor for authentication purposes



biometric characteristic

1/1 ?Why are phishing attacks so difficult to prevent ✓

Phishing is hard to stop because it tricks people into giving away info, and the fake
.messages look real. Even smart users can be fooled

1/1 Suppose that your boss asks you whether the company should require use ✓
.of password managers for all employees

Explain in a short paragraph why you might want everyone in the company
.to use a password manager

Using a password manager helps employees create and store strong, unique passwords for every account without having to remember them all. This reduces the risk of weak or reused passwords, which are common ways hackers break into systems. It also makes managing passwords easier and more secure, protecting the company's sensitive information from attacks like credential stuffing or phishing. Overall, requiring password managers improves
.security and saves time

Feedback

* ?How did you find the difficulty of this assignment

5 4 3 2 1
Too hard ☐ ☐ ☒ ☐ ☐ Too easy



* ?About how many MINUTES would you say you spent on this assignment
Just to set expectations for future students

30

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